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Spillovers of US interest rates: Monetary policy & information effects☆

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ABSTRACT

This paper quantifies the international spillovers of U.S. interest rates by accounting for the "Fed Response to News" channel. Using the identification strategy of Bauer and Swanson (2023a), we decompose monetary policy surprises into two components: a pure U.S. monetary policy shock and a "Fed Response to News" component around FOMC meetings. I find that a U.S. monetary tightening driven by pure policy shocks causes a global recessions, exchange rate depreciation, and tighter financial conditions. In contrast, a tightening driven by the "Fed Response to News" channel leads to an economic expansion, exchange rate appreciation, and looser financial conditions. Ignoring the "Fed Response to News" channel biases estimates, explaining recent atypical findings of expansionary impacts. By isolating these components, I reconcile traditional and recent views of monetary policy spillovers. Results are robust across advanced and emerging economies, alternative methods, and identification strategies.

1. Introduction

Understanding how US monetary policy affects other countries is central in international macroeconomics, reflecting the dollar's role as a global reserve currency and the Federal Reserve's broad influence on financial markets. Traditionally, US monetary tightening has been associated with economic contractions, currency depreciation, and financial distress internationally (Svensson and Van Wijnbergen, 1989; Obstfeld and Rogoff, 1995; Betts and Devereux, 2000). However, recent studies find an atypical response, where US interest rate hikes correspond to dollar depreciation and economic expansion abroad (Stavrakeva and Tang, 2019; Ilzetzki and Jin, 2021). In this paper, I show that the atypical dynamics documented in this recent literature can be explained by the "Fed Response to News" channel, in which incoming, publicly available economic news causes both the Federal Reserve and the private sector to revise its forecasts economic forecasts, see Bauer and Swanson (2023a). I argue that this effect biases conventional estimates, and by controlling for it, I restore the conventional view that US monetary tightening leads to a recession, a currency depreciation, and a global financial tightening.

Standard methods for identifying US monetary policy shocks consider monetary policy surprises around Federal Open Market Committee (FOMC) announcements (Gertler and Karadi, 2015; Nakamura and Steinsson, 2018a). However, recent literature argues

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that these surprises reflect both exogenous shocks and the private sector's updated beliefs about the state of the economy, referred to as the "Fed Information Effect" (see Nakamura and Steinsson 2018b, Jarocinski 2020), and updated beliefs about the FOMC's policy rule, referred to as the "Fed Response to News" channel (see Bauer and Swanson 2023a). Identifying pure monetary policy shocks from monetary policy surprises is essential for accurately studying their international spillovers. I address this using the recent identification strategy of Bauer and Swanson (2023a). By applying this identification strategy to both local projection and SVAR models, I show that pure policy shocks lead to traditional outcomes – recession, currency depreciation, and financial tightening – while the "Fed Response to News" channel component of monetary policy surprises is associated with an economic expansion and currency appreciation.

My results are robust for two alternative identification strategies that separate pure monetary policy shocks from information effects based on different methodologies and interpretations. A first alternative strategy is presented by Miranda-Agrippino and Ricco (2021), which argues the presence of a "signaling channel" of US monetary policy in monetary policy surprises.[1] A second alternative strategy is presented by Jarociński and Karadi (2020), which argues the presence of a systematic information disclosure shock that affects private sector beliefs about monetary policy and other economic fundamentals.[2] I show that my results are robust to these alternative identification strategies.[3] I interpret this result as evidence that purging monetary policy surprises of the "Fed Information Effect" is key to accurately estimating the international spillovers of US interest rates.

Finally, my main results are robust across various samples, including advanced economies (AEs) and emerging markets (EMEs), across time periods, geographic regions, and country characteristics such as current account and trade openness and exchange rate policy regimes. Results hold for alternative estimation methods in most countries when models are estimated country-by-country.

Related literature. This paper contributes to three main strands of literature. First, it builds on research that identifies and quantifies the international spillovers of US monetary policy shocks and their transmission channels. Much of this literature finds that a US monetary tightening is associated with negative spillovers abroad, such as economic recessions, currency depreciations, and tighter financial conditions. Key studies include Eichenbaum and Evans (1995) and Uribe and Yue (2006), which analyze data from the 1980s and 1990s, and Dedola et al. (2017), Vicondoa (2019), which extend the analysis to the late 2000s. Throughout this paper, I refer to these outcomes as the "conventional view" of a US monetary policy tightening. My contribution to this literature is threefold: (1) I introduce an identification scheme that removes informational content from US monetary policy decisions, finding that conventional results hold for data from the 2000s and 2010s; (2) I provide evidence that purging US monetary policy surprises of information effects is essential to estimate spillovers from pure monetary policy shocks accurately; and (3) I document that the magnitude of international spillovers may be increasing over time.

Second, this paper contributes to recent research documenting atypical dynamics in the response of advanced economies (AEs) and emerging market economies (EMEs) to US interest rate changes. For instance, Ilzetzki and Jin (2021) report a shift in the transmission of US policy shocks over recent decades, where US rate hikes are associated with a depreciation of the US dollar and an economic expansion abroad, a phenomenon the authors label a "puzzling" change in transmission. Similarly, Canova (2005) finds that US monetary tightening leads to currency appreciation in Latin America, contradicting the conventional view.[4] I contribute to this literature by presenting evidence that such atypical dynamics can be attributed to the "Fed Response to News" channel. Using multiple identification schemes, I show that the "Fed Response to News" channel component consistently explains atypical dynamics observed in recent studies while isolating pure monetary policy shocks reaffirms the conventional spillover effects. This result is also present when using alternative identification strategies that purge interest rate surprises from the "Fed Information Effect".[5]

Third, this paper relates to the literature studying US interest rate spillovers while controlling for informational effects around FOMC meetings. For example, Jarociński (2022) uses the identification strategy from Jarociński and Karadi (2020) to show that central bank information effects significantly influence transatlantic monetary policy spillovers, particularly in the co-movement of German and US government bond yields around ECB and FOMC announcements. Similarly, Degasperi et al. (2020) employs the (Miranda-Agrippino and Ricco, 2021) approach, which controls for "signaling information" effects around FOMC meetings.[6] My contribution to this strand is to specifically document spillovers resulting from the "Fed Response to News" channel. While studies like (Degasperi et al., 2020) explore transmission channels of US interest rates, they do not directly address the impact of informational effects. My findings suggest that changes in US interest rates driven by the "Fed Response to News" channel produce significant spillovers worldwide.[7]

[1] The authors show that monetary policy surprises are auto-correlated and correlated with the Federal Reserve's Greenbook forecasts. The authors interpret these empirical results as evidence of a "signaling channel" of US monetary policy, through which the FOMC conveys information about stronger-than-expected economic fundamentals to which it endogenously responds. Furthermore, the authors show that this "signaling channel" significantly biases the identification of conventional monetary policy shocks and its impact on the US economy.

[2] In addition, Cai et al. (2021) provide evidence on the information content of FOMC announcements by using text from both the Federal Reserve and New York Times articles to compute differences in expectations.

[3] In particular, I estimate the impulse response functions of pure monetary policy shocks and "Fed Information Effect" components following the methodologies introduced by Jarociński and Karadi (2020), Miranda-Agrippino and Ricco (2021), and Bu et al. (2021).

[4] See also Uribe and Yue (2006), which finds atypical dynamics in alternative identification schemes.

[5] Similarly, Camara (2023) documents biases from information effects of ECB interest rates in an international setting.

[6] Further examples include (Camara and Ramirez-Venegas, 2022), which uses the (Bauer and Swanson, 2023b) strategy to analyze transmission channels in Chile, and Camara et al. (2024).

[7] The potential impacts of the FOMC's information disclosure have also been examined theoretically, as in Ahmed et al. (2021).

2. Country sample, variables & methodology

In this section, I describe my dataset's construction, the local projection and SVAR methodologies used, and the identification strategies used across the paper.

Country sample. To comprehensively estimate international spillovers of US interest rates, I construct a sample of macroeconomic and financial variables for the most extensive possible period and number of countries. Specifically, the dataset spans from January 1988 to December 2021. It includes all countries with available data on five critical variables from the IMF or OECD: (i) nominal exchange rate against the US dollar, (ii) long-term interest rate, (iii) consumer price index, (iv) industrial production index, and (v) equity index. These variables capture key responses to US monetary policy and are essential for examining international spillovers. Long-term interest rates and equity indices are included as proxies for financial conditions, as higher interest rates do not always indicate tighter financial conditions in response to US rate changes (Hoek et al., 2022).

Following the IMF's classification, I group countries into Advanced Economies (AE) and Emerging Market Economies (EME). The sample comprises an unbalanced panel of 50 countries: 28 AEs and 22 EMEs. Compared to similar studies, this sample covers a broader range of countries and a longer timeframe. For example, Ilzetzki and Jin (2021) includes only 21 countries (12 AEs and 9 EMEs), Hoek et al. (2022) focuses on 20 EMEs, and Degasperi et al. (2020) uses 30 countries (15 AEs and 15 EMEs). My sample includes the same timeframe as (Ilzetzki and Jin, 2021), allowing for direct comparison.[8]

All variables are sourced from IMF and OECD datasets to ensure methodological consistency.[9] I exclude Euro Area countries once they adopt the euro, replacing them with a single Euro Area aggregate to prevent over-representation. Additionally, I show that results are robust when using a trade-weighted real exchange rate index, as real exchange rates often vary significantly across Euro Area countries despite a shared nominal rate.[10]

In Section 3, I demonstrate that results are robust across AE and EME samples and various time frames. Additionally, in Section 4, I show results are robust to different variable configurations, including additional macroeconomic, labor, trade, and financial indicators. Section 4.2 verifies results are present with the exact country and variable samples from Ilzetzki and Jin (2021) and other sub-samples based on the continent, exchange rate regime, and trade openness.

Identification strategies. The benchmark results in this paper are estimated using the identification strategy introduced by Bauer and Swanson (2023b). The authors construct monetary policy surprises by taking the first principal component of changes in the first four quarterly Eurodollar futures contracts around FOMC announcements. To isolate pure monetary policy shocks, they orthogonalize these surprises from macroeconomic and financial information released around the FOMC meeting:

$$mps_t = \alpha + \beta X_t + u_t \tag{1}$$

where mps_t is the monetary policy surprise, X_t denotes a vector of controls and u_t is the regression residual.[11,12] The pure monetary policy (MP) component of monetary policy surprises is defined as:

$$\text{MP component} = mps_t - \hat{\alpha} - \hat{\beta} X_t \tag{2}$$

where $\hat{\alpha}, \hat{\beta}, X_t$ correspond to the estimated coefficients and predictors from Eq. (1). I define the "Fed Response to News" component as:

$$\text{RN component} = \hat{\alpha} + \hat{\beta} X_t \tag{3}$$

This method is similar to that of Miranda-Agrippino and Ricco (2021), which isolates the Fed's "signaling channel".

This identification strategy has two main advantages. First, it allows for a longer time sample (February 1988 to December 2019), which is especially valuable given that US monetary policy has become more systematic over time, making monetary policy surprises smaller and less informative (Ramey, 2016). Second, Bauer and Swanson (2023b) provides evidence supporting their interpretation of the "Fed Response to News" channel over previous interpretations.[13]

[8] For robustness, Section 4 shows that results hold across different sub-samples in both country composition and timeframe.

[9] Data from IMF's International Financial Statistics and OECD Statistics is supplemented with constructed long-term interest rates where needed by combining a country's EMBI Spread and Kenneth French's short-term US risk-free rate (Camara et al., 2024). Details are in Appendix A.

[10] The real exchange rate index follows (Darvas, 2012) and is available at https://www.bruegel.org/publications/datasets/real-effective-exchange-rates-for-178-countries-a-new-database. The index is normalized such that an increase indicates home currency depreciation. Persistent real exchange rate differences within the Euro Area are documented by Crucini et al. (2005) and Fidora et al. (2021).

[11] This equation relates to equation 14 from (Bauer and Swanson, 2023b).

[12] The vector of controls are (i) Non-farm payrolls surprise: the surprise component of the most recent non-farm payrolls release prior to the FOMC announcement, measured as the difference between the released value of the statistic minus the median expectation for that release from the Money Market Services survey; (ii) Employment growth: the log change in nonfarm payroll employment from 1 year earlier to the most recent release before the FOMC announcement; (iii) S&P 500: the log change in the S&P 500 stock market index from 3 months (65 trading days) before the FOMC announcement to the day before the FOMC announcement; (iv) Yield curve slope: the change in the slope of the yield curve from 3 months before the FOMC announcement to the day before the FOMC announcement, measured as the second principal component of 1-to −10-year zero-coupon Treasury yields; (v) Commodity prices: the log change in the Bloomberg Commodity Spot Price index from 3 months before the FOMC announcement to the day before the FOMC announcement; (vi) Treasury skewness: the implied skewness of the 10-year Treasury yield, measured using options on 10-year Treasury note futures with expirations in 1–3 months, averaged over the preceding month, from Bauer and Chernov (2024).

[13] Acosta (2022), however, offers evidence countering (Bauer and Swanson, 2023a) and emphasizes the role of information disclosure and misperceptions around FOMC meetings.

To confirm the robustness of the benchmark results, I also apply alternative identification strategies that provide different interpretations of the information effects around FOMC announcements. Specifically, I estimate international spillovers using methodologies from Jarociński and Karadi (2020), and Miranda-Agrippino and Ricco (2021) each of which highlights the Fed's role in conveying information on the US economy around FOMC meetings. I also use (Bu et al., 2021), which employs a Fama–MacBeth two-step procedure to isolate a pure monetary policy shock and find the results robust. Additionally, I replicate recent results from Ilzetzki and Jin (2021) using the methodology of Nakamura and Steinsson (2018a), emphasizing the significance of information effects in US monetary transmission.

This paper does not aim to evaluate the validity of each identification method. Instead, it highlights that, regardless of interpretation or methodology, controlling for the "Fed Response to News" or other information effects around FOMC announcements is crucial to accurately measure the international spillovers of US interest rates.

Methodology. I quantify the international spillovers of US interest rates by estimating both local projection regressions á la (Jordà, 2005) and Bayesian SVAR models.

First, local projection regressions allow me to estimate the impact of the two components of monetary policy surprises for large, unbalanced panels of countries. This exercise allows me to estimate the international spillovers of US interest rates for the largest possible sample in terms of the time and country dimensions. The main local projection regression estimated is given by

$$y_{i,t+h} = \beta_h^{MP} i_t^{\mathrm{MP}} + \beta_h^{RN} i_t^{\mathrm{RN}} + \sum_{j=1}^{J_y} \delta_i^j y_{i,t-j} + \sum_{j=1}^{J_x} \alpha_i^j x_{i,t-j} + \sum_{j=1}^{J_i} \left(\phi_i^j i_{t-j}^{\mathrm{MP}} + \varphi_i^j i_{t-j}^{\mathrm{RN}} \right) + \epsilon_{i,t} \tag{4}$$

where $y_{i,t+h}$ represents the outcome variable from country i's at time horizon of h months from date t. Coefficients β_h^{MP} and β_h^{RN} give the impulse response of outcome variable y at horizon h of a pure monetary policy shock (MP) component and of the "Fed Response to News" (RN) component, respectively. The specification includes J_y lags of the outcome variable, y_i, J_x lags of the other "country-specific" variables x_i and J_i lags of the monetary policy surprises components. For the benchmark sample, I set the lags on the dependent and other control variables at 2, i.e., $J_y = J_x = 2$, and the number of lags on the components of monetary policy components at 4, i.e., $J_i = 4$. Introducing lagged values of the dependent variable controls for any potential seasonality in the monthly variables. I choose two lags considering the Bayesian Information Criterion (BIC) and the Hannan-Quinn (HQ) information criteria. I also introduce lagged values of the monetary policy components for efficiency reasons as suggested by Jordà (2023).[14] Additionally, I estimate an alternative specification of Eq. (4) in which I follow the standard high-frequency identification assumption ("Standard HFI") and replace the MP and RN components with the monetary policy surprise. The standard errors are clustered at the time level, which incorporates the fact that all countries are hit by the shocks simultaneously. In Section 4.2, I show that all of the benchmark results are robust to alternative specifications of Eq. (4), which include time trend and country fixed effects.

Second, I estimate Bayesian SVAR models to study potential sources of heterogeneity across countries and to provide a robustness check for my local projection regression analysis. First, I estimate a country-by-country VAR model, introducing the MP and RN components as exogenous variables. Bayesian methods permit me to estimate the model for each country separately and study potentially heterogeneous impacts of US interest rates across countries, even with a limited time series. The results show that they are prevalent across most countries in my sample and are not driven by outliers. Second, I estimate several pooled and mean-group Bayesian panel SVAR model specifications. Given the well-known bias–variance trade-off between the two methodologies, these exercises provide a robustness check to the local projection regression result, see Plagborg-Møller and Wolf (2021). The details of these Bayesian SVAR models are described in Appendix B.

3. Spillovers of monetary policy & information effects

This section presents the main results, quantifying the impact of the two components of monetary policy surprises: the pure monetary policy shock (MP) and the "Fed Response to News" (RN) component. I compare these impulse response functions with those derived from the standard high-frequency identification ("Standard HFI") approach, which uses only unadjusted monetary policy surprises. Two main results emerge: (1) the MP and RN components produce opposite international spillovers, and (2) the RN component explains the atypical dynamics observed in recent studies that rely on the Standard HFI approach.

I begin by assessing the importance of decomposing US monetary policy surprises into MP and RN components for identifying international spillovers. The first two columns of Fig. 1 present the impulse response functions of the macroeconomic and financial variables to an MP and RN component of monetary policy surprises. Comparing the responses in these columns reveals that controlling for the "Fed Response to News" (RN) channel separates monetary policy surprises into two components with nearly opposite effects. For example, while the MP component leads to exchange rate depreciation and a decline in industrial production, the RN component induces exchange rate appreciation and an increase in industrial output. This distinction underscores the importance of identifying pure monetary policy shocks for accurately quantifying the international spillovers of US interest rates.

Next, I describe in greater detail the impulse response functions. Panel (1.1) shows a significant depreciation of the nominal exchange rate on impact, followed by further depreciation over the first year, with statistical significance at the 90% level for the first four months. Panel (2.1) shows a 100 basis-point rise in long-term interest rates on impact, peaking at 200 basis points two months after the shock, and remaining statistically significant at the 90% level for a year. Panel (3.1) shows a gradual rise in

[14] Results are robust to introducing several alternative lag structures of dependent variables and components of monetary policy surprises.

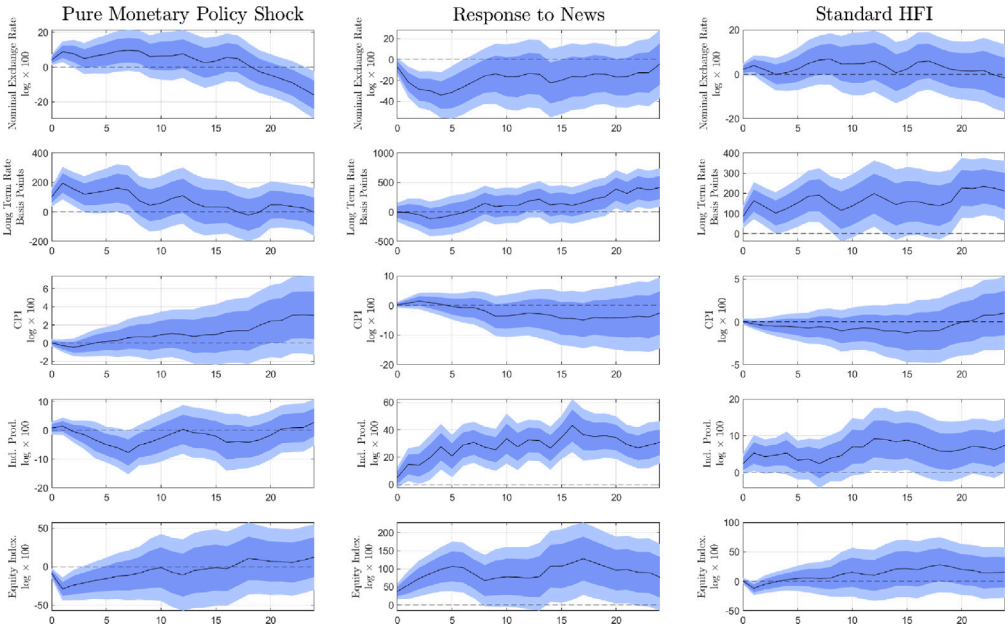


Fig. 1. Impulse response functions
Benchmark specification.
Note: The figure is comprised of 15 sub-figures ordered in three columns and five rows. The left column relates to the estimates of β^{MP} in Eq. (4), the middle column relates to the estimate of β^{RN} in Eq. (4), while the right column relates to estimating Eq. (4), replacing the MP and RN components with the un-orthogonalized monetary policy surprise. The rows represent the impact on (i) the nominal exchange rate with the US dollar (in logs times 100), (ii) long-term interest rates in basis points, (iii) the consumer price index (in logs times 100), (iv) the industrial production index (in logs times 100), (v) the equity index (in logs times 100). The solid black line represents the point estimate, the dark blue area represents the 68% confidence interval, and the light blue area represents the 90% confidence interval. In the text, when referring to Panel (i, j), i refers to the row and j to the column of the figure. Each variable, in its own transformation, is demeaned at the country level.

consumer prices, statistically significant at the 68% level after one year. Panel (4.1) indicates a recessionary effect, with industrial production reaching its trough 7–8 months after the shock. Finally, Panel (5.1) shows a substantial drop in equity indexes, with an initial 10% fall and a trough of 25% within two months, not returning to pre-shock levels until a year later.[15]

The spillovers of the RN component of monetary policy surprises are opposite to those of the MP component. Panel (1.2) reveals a hump-shaped, statistically significant appreciation of the exchange rate, persisting up to 18 months. Panel (2.2) shows long-term interest rates with minimal immediate response but a sustained rise between 6–9 months after the shock. Panel (3.2) displays an initial increase in the consumer price index, later reversing below pre-shock levels after nine months. Panel (4.2) shows a significant expansion in industrial production, reaching a peak 16–18 months post-shock. Lastly, Panel (5.2) illustrates a substantial rise in equity indexes, remaining significantly above pre-shock levels for 24 months. This comparison highlights that MP and RN components produce contrasting responses across macroeconomic and financial variables.

Lastly, I argue that following the "Standard HFI" strategy may lead to biased impulse response functions. I modify the benchmark specification in Eq. (4) by replacing the MP and RN components with the un-orthogonalized monetary policy surprise. The third column of Fig. 1 exhibits the impulse response functions under this identification strategy. Across the different variables, the impulse responses are an average of the responses presented for the MP and FIE shocks in the left and middle columns. Most shockingly, the "Standard HFI" strategy leads to a qualitatively different response of the industrial production index from that arising after an MP shock. Panel (3.4) shows that under the "Standard HFI" strategy, the industrial production increases significantly after a monetary policy surprise. This expansion is statistically different from zero on impact and remains persistently above pre-shock levels for more than 12 months after the initial shock. Similarly, after exhibiting a shortfall on impact, the response of the equity index turns positive and remains above pre-shock levels 9 months after the initial shock. In addition to these qualitative differences, there are notable quantitative differences in the impulse response functions of other variables. Panel (3.1) shows that under the "Standard HFI" strategy, the depreciation of the nominal exchange rate is not statistically different from zero. Consequently, following the "Standard HFI" strategy may lead to underestimating the depreciation of the exchange rate after a pure US monetary policy shock.

Results replacing the NER with a REER. To test the robustness of the main findings, I re-estimate the model using the multilateral trade-weighted real exchange rate index in place of the nominal exchange rate, as discussed in Section 2. Fig. 2 presents the results

[15] The equity index is expressed in domestic currency. Thus, the decline is not driven by the exchange rate depreciation. This suggests a real decline in equity value.

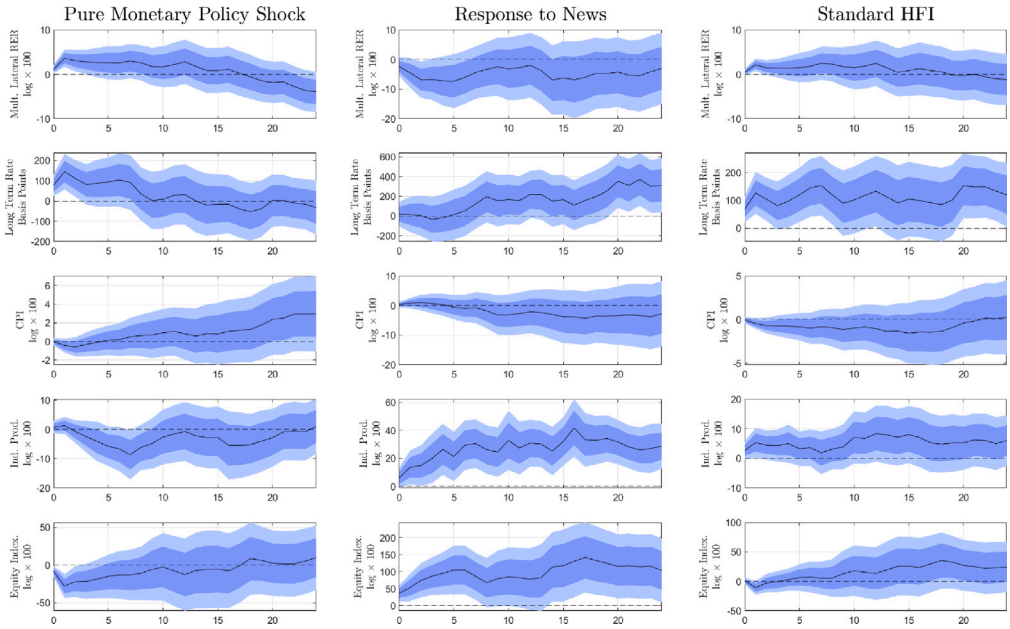


Fig. 2. Impulse response functions
Multilateral real exchange rate specification.
Note: The figure is comprised of 15 sub-figures ordered in three columns and five rows. The left column relates to the estimates of β^{MP} in Eq. (4), the middle column relates to the estimate of β^{RN} in Eq. (4), while the right column relates to estimating Eq. (4), replacing the MP and RN components with the un-orthogonalized monetary policy surprise. The rows represent the impact on (i) the multilateral trade-weighted real exchange rate index (in logs times 100), (ii) long-term interest rates in basis points, (iii) the consumer price index (in logs times 100), (iv) the industrial production index (in logs times 100), (v) the equity index (in logs times 100). The solid black line represents the point estimate, the dark blue area represents the 68% confidence interval, and the light blue area represents the 90% confidence interval. In the text, when referring to Panel (i, j), i refers to the row and j to the column of the figure. Each variable, in its own transformation, is demeaned at the country level.

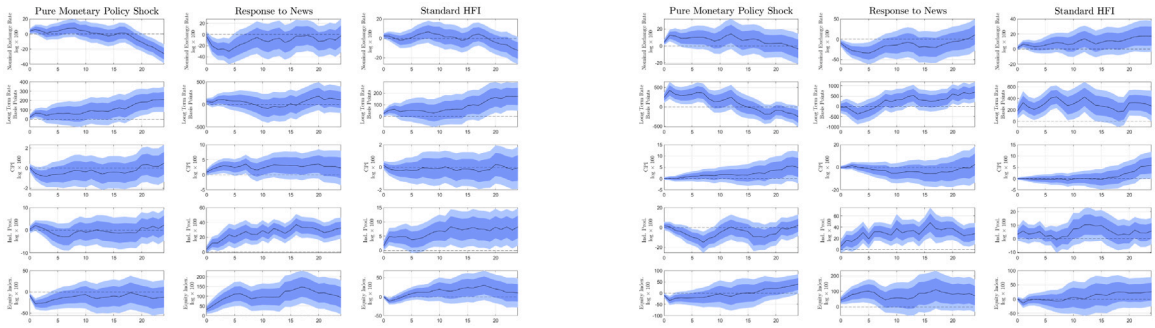
under the REER specification. Panel (1.1) shows that in response to the MP component, countries' real exchange rate depreciates, consistent with the conventional view from a contractionary monetary policy shock. Conversely, Panel (1.2) demonstrates that the RN component leads to a significant appreciation of the real exchange rate, further confirming the opposite directions of these "Fed Response to News" channel. Panel (1.3) shows that, under the "Standard HFI" approach, the real exchange rate response is muted and not statistically different from zero. The responses of the other macroeconomic and financial variables, shown in the remaining rows of Fig. 2, closely resemble those in Fig. 1, both qualitatively and quantitatively. This consistency across specifications reinforces the importance of decomposing US monetary policy surprises into MP and RN components to fully understand and accurately capture international spillovers of US interest rates.

Results across different country & time-samples. To validate the robustness of the main results, I examine sub-samples divided by country classification and time period. This approach assesses whether the identified spillovers hold consistently across Advanced and Emerging Market Economies and over different time periods.

I begin by estimating the impulse response functions for the MP and RN components and following the "Standard HFI" for the panel of Advanced Economies (Fig. 3(a)) and for the panel of Emerging Market Economies (Fig. 3(b)), separately.

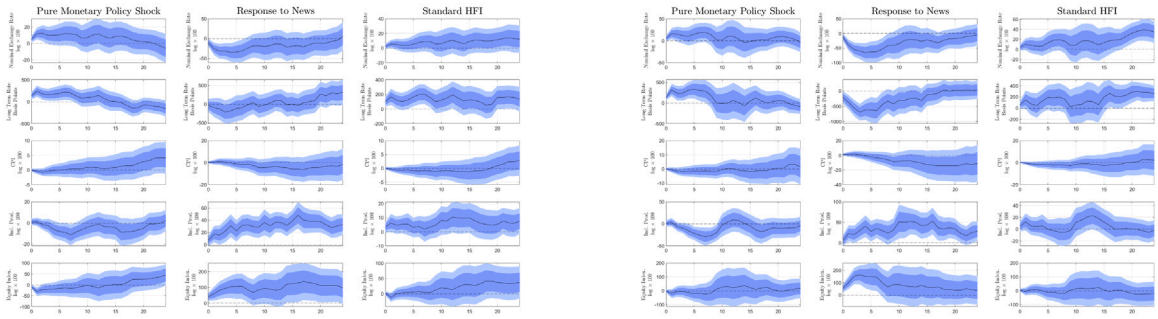
Across the two sub-samples, the main results presented in Fig. 1 hold, with the two components of monetary policy surprises leading to opposite dynamics and the "Standard HFI" leading to a weighted average of them. Still, there are some noteworthy quantitative differences. Notably, the consumer price index response differs across these groups: EMEs show a persistent increase following an MP component shock, while AEs experience a significant decline. This finding aligns with literature (e.g., García-Cicco and García-Schmidt 2020 and Auclert et al. 2021) emphasizing EMEs' higher reliance on imported goods, leading to greater sensitivity to exchange rate depreciation. Additionally, the MP component's impact on long-term interest rates is both faster and greater in magnitude in EMEs, likely reflecting these economies' smaller, less developed financial markets and greater reliance on international capital flows (see Dages et al. 2000, Broner et al. 2013, Cortina et al. 2018, Abraham et al. 2020). Figures 5 and 6 in Appendix C present this robustness check for samples that replace the nominal exchange rate with the multilateral exchange rate.

Lastly, I estimate the impulse response functions for the MP and RN components and following the"Standard HFI" for a sample starting in January 1998 and for a sample starting in January 2008, 10 and 20 years after the start of the benchmark sample, respectively. This exercise allows me to test whether the international spillovers of US interests have varied significantly in recent years as suggested by Ilzetzki and Jin (2021). The results are presented in Fig. 4. The resulting impulse response functions in the two time sub-samples are in line with those obtained for the benchmark sample presented in Fig. 1. One quantitative difference worth mentioning is that the impact of the MP component on industrial production seems to have increased across time. For the



(a) Advanced Economies (b) Emerging Market Economies

Fig. 3. Impulse Response functions
Separate samples for adv. & emerging economies.
Note: Each of the two figures are comprised of 15 sub-figures ordered in three columns and five rows. The first figure to the left is constructed by estimating Eq. (4) for the sample of Advanced Economies. The second figure to the right is constructed by estimating Eq. (4) for the sample of Emerging Market Economies. See Appendix A for the list of countries in each sample. The left column relates to the estimates of β^{MP} in Eq. (4), the middle column relates to the estimate of β^{RN} in Eq. (4), while the right column relates to estimating Eq. (4), replacing the MP and RN components with the un-orthogonalized monetary policy surprise. The rows represent the impact on (i) the nominal exchange rate with respect to the US dollar (in logs times 100), (ii) long-term interest rates in basis points, (iii) the consumer price index (in logs times 100), (iv) the industrial production index (in logs times 100), (v) the equity index (in logs times 100). The solid black line represents the point estimate, the dark blue area represents the 68% confidence interval, and the light blue area represents the 90% confidence interval. In the text, when referring to Panel (i, j), i refers to the row and j to the column of the figure. Each variable, in its own transformation, is demeaned at the country level.



(a) Sample January 1998 - Onward (b) Sample January 2008 - Onward

Fig. 4. Impulse response functions
Samples starting in January 1998 & January 2008.
Note: Each of the two figures are comprised of 15 sub-figures ordered in three columns and five rows. The first figure to the left is constructed by estimating Eq. (4) for the sample starting in January 1998. The second figure to the right is constructed by estimating Eq. (4) for the sample starting in January 2008. The left column relates to the estimates of β^{MP} in Eq. (4), the middle column relates to the estimate of β^{RN} in Eq. (4), while the right column relates to estimating Eq. (4), replacing the MP and RN components with the un-orthogonalized monetary policy surprise. The rows represent the impact on (i) the nominal exchange rate with respect to the US dollar (in logs times 100), (ii) long-term interest rates in basis points, (iii) the consumer price index (in logs times 100), (iv) the industrial production index (in logs times 100), (v) the equity index (in logs times 100). The solid black line represents the point estimate, the dark blue area represents the 68% confidence interval, and the light blue area represents the 90% confidence interval. In the text, when referring to Panel (i, j), i refers to the row and j to the column of the figure. Each variable, in its own transformation, is demeaned at the country level.

benchmark sample, Fig. 1 shows a drop in industrial production below 10%, while Fig. 4 shows an impact close to 12% and to 20% for the sample starting in 1998 and 2008 respectively. Figures 7 and 9 in Appendix C repeats the exercise for the sample of AE and EME respectively to show that this trend is present across countries.[16]

To summarize, by introducing an identification scheme that deconstructs US monetary policy surprises into two distinct components, I demonstrate that an increase in US interest rates can produce markedly different international spillovers. While a pure monetary policy (MP) shock results in conventional outcomes, a monetary policy surprise driven by the "Fed Response to News"(RN) channel induces economic expansion, exchange rate appreciation, and looser financial conditions. I further argued that

[16] Additionally, Figures 8 and 10 in Appendix C show that this result is also present when using the sample that replaces the nominal exchange rate with the multilateral real exchange rate.

relying on a 'Standard HFI' strategy underestimates the spillovers of a US monetary policy shock and leads to atypical dynamics, such as economic expansion. Finally, I show that these findings hold for both sub-samples of Advanced and Emerging Market Economies, with only minor variations, and across different time samples.

4. Additional results & robustness checks

This section provides additional results and robustness checks that complement and validate the findings presented in Section 3. In Section 4.1, I present three key additional results: (i) I estimate the international spillovers of US interest rates using samples with alternative variable selections; (ii) I demonstrate that results remain qualitatively similar when using alternative identification schemes to account for the "Fed Information Effect" under different methodologies and interpretations; and (iii) I estimate the international spillovers on a country-by-country basis using a Structural Vector Autoregression (SVAR) model. In Section 4.2 I confirm the robustness of my main findings by testing alternative econometric specifications and performing a series of sub-sample analyses. The corresponding figures and detailed results are provided in Appendix C.

4.1. Additional results

4.1.1. Additional variables

First, I extend the benchmark sample with additional variables to provide supporting evidence that the MP and RN components of US monetary policy shocks have distinct international spillovers. Figures 11, 12, 13 and 14 in Appendix C describe the impact of US interest rate changes on additional variables capturing countries' financial conditions and economic performance.

Financial conditions. Figure 11 illustrates the pass-through of US interest rate shocks to domestic monetary policy and short-term private sector lending rates. The left column shows that the MP component results in an immediate tightening of financial conditions, with sharp increases in both policy and lending rates. However, this effect is temporary, as rates quickly return to pre-shock levels. In contrast, the middle column demonstrates that the RN component produces a delayed, gradual, and more persistent increase in domestic rates, with significant effects emerging between six and nine months after the initial shock. These dynamics align closely with the behavior of long-term interest rates in the benchmark sample. Using the "Standard HFI" approach, however, one might incorrectly conclude that a US interest rate hike consistently raises policy rates for up to two years post-shock.

Additional real economic variables. Figure 12 examines the impact of US interest rates into three additional real economic variables: (i) a construction index, (ii) a mining index, and (iii) a retail sales index. Although these variables generally have lower coverage than industrial production indices, they represent shares of GDP comparable to or larger than manufacturing. The left column reveals that a monetary tightening due to the MP component leads to a persistent decline in construction, mining, and retail activity. Conversely, the middle column shows that RN component shocks significantly boost these indices, signaling economic expansion. The right column illustrates how the "Standard HFI" approach could erroneously suggest that US monetary tightening promotes economic expansion, consistent with recent findings in the literature.

Labor market impact. Figure 13 explores the effects of US monetary policy shocks on unemployment rates. The MP component (left panel) triggers a hump-shaped increase in unemployment, peaking within a few quarters post-shock. Conversely, the RN component (middle panel) results in a persistent decline in unemployment for up to 18 months. Under the "Standard HFI" approach, a US monetary tightening appears to reduce unemployment, suggesting positive international spillovers, a conclusion at odds with underlying dynamics.

International Trade Variables. Figure 14 shows the impact of US interest rates on international trade variables (i) a country-specific commodity export price index, (ii) a country-specific commodity import price index, (iii) total export values (FOB, USD), and (iv) total import values (CIF, USD). The left column shows that MP component shocks reduce international trade, evidenced by falling commodity price indices and declines in total traded values. In contrast, the RN component (middle column) expands trade through higher commodity prices and increased trade values. The right column, consistent with prior analyses, demonstrates that the "Standard HFI" approach might incorrectly attribute rising commodity prices and ambiguous trade impacts to US monetary tightening.

Robustness in variable selection. To further test robustness, I simplify the benchmark specification by removing key variables. While reducing variables increases the sample size – adding more countries and observations per country – it also heightens the risk of model misspecification. I estimate impulse response functions for a sample that removes the long-term interest rate (see Figures 19 to 24 in Appendix C), and for a sample that removes both the long term interest rate and the equity index (see Figures 25 to 30 in Appendix C).[17] The estimated impulse response functions are in line with those presented in Section 3, though confidence intervals are marginally wider.

Overall, extending the benchmark sample with additional variables corroborates the main findings presented in Section 3. The MP component of US monetary policy surprises leads to an immediate deterioration of financial conditions – evidenced by higher domestic interest rates – and causes persistent declines in economic activity, trade, and increases in unemployment. By contrast, the RN component leads to no immediate changes in domestic rates and instead drives significant expansions in economic output, trade, and reductions in unemployment. Figures 15 to 18 in Appendix C confirm that these results remain robust when using a sample that substitutes nominal exchange rates with trade-weighted real exchange rates.

[17] Removing the long-term interest and/or the equity index does not increase significantly the number of countries in the sample, from 50 to 58.

4.1.2. Alternative interpretations for the "fed information effect"

Second, I demonstrate that alternative identification strategies, which control for the "Fed Information Effect", yield results consistent with those presented in Section 3. I employ the identification approaches developed by Bu et al. (2021), Jarociński and Karadi (2020) and Miranda-Agrippino and Ricco (2021), each offering distinct methodologies and interpretations for addressing the "Fed Information Effect".

Identification Strategy by Bu et al. (2021)**.** Using the method proposed by Bu et al. (2021), I estimate the international spillovers of US interest rates with a monetary policy shock extracted through a Fama–MacBeth two-step procedure. This approach isolates a pure monetary policy shock, analogous to the MP component in my benchmark analysis, for the period January 1994 to September 2019. The authors argue that this shock is largely unpredictable and free from the "Fed Information Effect".

Figure 31 in Appendix C present the results of estimating the impulse response functions by replacing the MP and FIE components in Eq. (4) with Bu et al. (2021)'s shock. This monetary policy shock purged of the "Fed Information Effect" leads to an exchange rate depreciation, a persistent increase in long-term rates, and a drop in both industrial production and equity indexes.[18] This results are aligned with the impact of the MP component presented in Section 3.

Identification Strategy by Jarociński and Karadi (2020)**.** identify two components of US monetary policy surprises by imposing sign-restriction moment conditions on the high-frequency surprises of monetary policy surprises and the stock market around FOMC announcements. On the one hand, a pure monetary policy shock is identified as a structural shock that creates a negative co-movement between monetary policy surprises and the stock market. On the other hand, an information disclosure shock by the FOMC about the state of the economy is a second structural shock that creates a positive co-movement between monetary policy surprises and the stock market.

Figures 33 and 34 in Appendix C show results for the pure monetary policy and information disclosure shocks, respectively. Comparing these with the left and middle columns of Fig. 1, I find qualitatively similar spillovers. The pure monetary policy shock corresponds to a nominal exchange rate depreciation, higher long-term interest rates, and lower equity indices, though its impact on industrial production is muted for the full sample. Notably, for the post-2008 sample, industrial production declines significantly. Similarly, Jarociński and Karadi (2020)'s information disclosure shock aligns with the impact of the RN component, showing an exchange rate appreciation, no initial impact on long-term rates, and expansions in both industrial production and equity indices.[19]

Identification Strategy by Miranda-Agrippino and Ricco (2021)**.** Miranda-Agrippino and Ricco (2021) identify two components of US monetary policy surprises by projecting them onto their own lags and the Federal Reserve's information set (Greenbook forecasts) for the period January 1991 to December 2015. The orthogonalized component is interpreted as a pure monetary policy shock, while the predictable component captures the "signaling effect".

Figures 37 and 38 in Appendix C present the results for these two components. The pure monetary policy component leads to a nominal exchange rate depreciation, higher long-term interest rates, and declines in both industrial production and equity indices, consistent with my benchmark results on the MP component in Fig. 1. Meanwhile, the "signaling effect" component mirrors the RN component in Fig. 1, causing an exchange rate appreciation, an initial drop in long-term rates, and expansions in both industrial production and equity indices.[20]

Across these three alternative identification strategies that control for the "Fed Information Effect" I find international spillovers of US interest rates consistent with the results presented in Section 3. Pure monetary policy shocks lead to nominal exchange rate depreciation, tighter financial conditions, and declines in industrial production. In contrast, measures capturing the "Fed Information Effect" produce nominal exchange rate appreciation, looser financial conditions, and expansions in industrial production. These findings provide strong evidence that controlling for information effects around FOMC meetings is essential for accurately estimating the international spillovers of US interest rates.

4.1.3. Robustness to country-by-country SVAR models

Thus far, the impulse response functions have been estimated using the panel structure of the dataset. While this approach provides an average impact across countries, it does not account for potential country-specific heterogeneity. To address this limitation, I estimate a series of country-by-country Bayesian Structural Vector Autoregression (SVAR) models. Bayesian SVARs are particularly well-suited for this analysis, given the limited number of observations available for many countries in the sample.

The Bayesian SVAR models employ standard Minnesota priors, with parameter choices informed by the literature.[21] See Appendix B of the Appendix for additional details on this model. Each model is estimated for the longest available time sample for each country and incorporates the same country-specific variables as in the benchmark specification, along with a set of US-specific variables: (i) the two year treasury rate, (ii) the Federal Reserve's real trade-weighted dollar index, (iii) the PCE price index, (iv) the dollar value of US exports, (v) the dollar value of US imports, (vi) the excess bond premium by Gilchrist and Zakrajšek (2012), and (vii)

[18] Figure 32 in Appendix C shows that these results are also present for the sample that replaces the nominal exchange rate with the trade-weighted multilateral real exchange rate.

[19] Figures 35 and 36 in Appendix C show that these findings also hold for trade-weighted real exchange rates sample.

[20] Figures 39 and 40 in Appendix C show that these results are also present for the sample which replaces the nominal exchange rate with the trade-weighted multilateral real exchange rate.

[21] The model is specified with 12 lags, an auto-regressive coefficient of 0.8, an overall tightness parameter of 0.1, a cross-variable weighting parameter of 0.5, and a lag decay parameter of 1. Exogenous variables, such as the MP and RN components, are assigned a prior with a mean of 0 and variance of 100. These parameter values are standard in the literature, and altering them does not result in substantive changes to the results.

industrial production index. The inclusion of US-specific variables in country-by-country SVAR models is standard practice in the literature, as it reduces the risk of model misspecification.[22] The MP and RN components are introduced as exogenous variables, one at a time, to ensure clear identification of their distinct impacts. This country-specific approach provides further insights into the heterogeneity of international spillovers while maintaining comparability with the benchmark results.

Figures 41 to 50 in Appendix C present the impulse response functions for the five benchmark variables for both AE and EMEs. On the one hand, in response to the MP component of monetary policy surprises, 90% of the countries exhibited a depreciation of the nominal exchange rate (see Figures 41 and 46), 82.5% of the countries exhibited a fall in industrial production (see Figures 42 and 47), and 92.5% of the countries exhibited a fall in the equity index, respectively (see Figures 44 and 49).[23,24,25] On the other hand, in response to the RN component of monetary policy surprises, 95% of the countries experience an appreciation of the nominal exchange rate (see Figures 41 and 46), 95% of the countries experience an expansion of industrial production (see Figures 42 and 47), and 97.5% of the countries experience an expansion of the equity index (see Figures 44 and 49).[26,27,28]

To further highlight these results, Figure 51 in Appendix C pools the median impulse response functions across countries and plots the interquartile range for both the MP and RN components. The pooled results confirm that, while there is heterogeneity in response magnitude across countries, the vast majority of countries exhibit patterns consistent with the benchmark results presented in Section 3.

These findings reinforce the conclusion that the MP and RN components of monetary policy surprises generate opposing international spillovers. The MP component generally leads to tighter financial conditions, reduced industrial production, and lower equity indices, while the RN component produces looser financial conditions, expanded industrial production, and higher equity indices. These results underscore the importance of controlling for the "Fed Response to News" channel to accurately estimate the global impact of US interest rates.[29]

4.2. Robustness checks

This section presents a series of robustness checks to demonstrate that the main results in Section 3 are robust across alternative econometric specifications, panel SVAR results, and different sub-sample exercises.

4.2.1. Alternative econometric specifications

Alternative specifications to Eq. (4)**.** To verify the robustness of the results in Section 3, I estimate the impulse response functions using two alternative econometric specifications:

$$y_{i,t+h} = \beta_h^{MP} i_t^{\mathrm{MP}} + \beta_h^{RN} i_t^{\mathrm{RN}} + \\ + \sum_{j=1}^{J_y} \delta_i^j y_{i,t-j} + \sum_{j=1}^{J_x} \alpha_i^j x_{i,t-j} + \sum_{j=1}^{J_i} \left(\phi_i^j i_{t-j}^{\mathrm{MP}} + \varphi_i^j i_{t-j}^{\mathrm{RN}} \right) + \mu_1 \times t + \mu_2 \times t^2 + \epsilon_{i,t} \tag{5}$$

$$y_{i,t+h} = \beta_h^{MP} i_t^{\mathrm{MP}} + \beta_h^{RN} i_t^{\mathrm{RN}} + \\ + \sum_{j=1}^{J_y} \delta_i^j y_{i,t-j} + \sum_{j=1}^{J_x} \alpha_i^j x_{i,t-j} + \sum_{j=1}^{J_i} \left(\phi_i^j i_{t-j}^{\mathrm{MP}} + \varphi_i^j i_{t-j}^{\mathrm{RN}} \right) + \mu_1 \times t + \mu_2 + \phi_i \times t^2 + \Gamma_i + \epsilon_{i,t} \tag{6}$$

Eq. (5) includes a linear and a quadratic time trend. Eq. (6) adds country fixed effects.[30] Figures 63 and 64 in Appendix C show that the results remain robust under these alternative specifications.[31]

Panel SVAR econometric specification. I conduct additional robustness checks by estimating impulse response functions using two-panel SVAR models: (i) a pooled model and (ii) a mean-group model. Details on these models are found in Appendix B. Analogous

[22] See Degasperi et al. (2020) and Camara et al. (2024). The selected US variables align with standard practices in the literature examining US monetary policy effects, such as Bauer and Swanson (2023a). For details on each country's sample start and end dates, refer to Appendix A.

[23] The countries that exhibited either an exchange appreciation or a non-significant response are China, Finland, India, Venezuela, and Ukraine. Out of the 40 countries in this sample, only 4 did not exhibit an exchange rate depreciation.

[24] The countries that exhibited either an industrial production expansion or a non-significant response are Denmark, Poland, Slovenia, Switzerland, UK, Croatia and Romania. Out of the 40 countries in this sample, only 7 did not exhibit a drop in industrial production.

[25] The countries that exhibited either an equity index expansion or a non-significant response are Slovenia, India, and Pakistan. Out of the 40 countries in this sample, only 3 did not exhibit a drop in the equity index.

[26] The countries that exhibited either an exchange depreciation or a non-significant response are Finland and Japan. Out of the 40 countries in this sample, only 2 did not exhibit an exchange rate appreciation.

[27] The countries that exhibited either an industrial production fall or a non-significant response are Norway and Iceland. Out of the 40 countries in this sample, only 2 did not exhibit an increase in industrial production.

[28] The only country that exhibited either an equity index fall or a non-significant response is Slovenia. Out of the 40 countries in this sample, only 1 did not exhibit an increase in the equity index.

[29] Figures 52 to 62 repeat the country by country exercise for the sample that replaces the nominal exchange rate with the trade-weighted multilateral real exchange rate. All the results are robust.

[30] Note that the specification given by Eq. (6) introduces both lagged values of the independent variable and country fixed effects. As shown by Nickell (1981), this could lead to biased estimates. However, I also estimate the impulse response functions using this specification as it was the method followed by Ilzetzki and Jin (2021).

[31] Figures 63 and 64 in Appendix C show that the results remain robust under these alternative specifications in the sample with the trade-weighted multilateral real exchange rate.

to the country-by-country estimated in the previous section, I introduce the MP and RN components and the complete monetary policy surprise as exogenous variables.[32]

For the pooled panel SVAR model, I estimate impulse response functions for samples starting in January 1988, January 1998, and January 2008 and present them in Figures 67 and 69 in Appendix C. In these figures, the MP and RN components produce opposite international spillovers, consistent with the main results in Section 3.[33] For the mean-group model, I average demeaned variables across countries and estimate a one-unit SVAR on the aggregated data. Again, the results confirm the robustness of the main findings, as shown in Figures 73 to 75 in Appendix C.[34]

Results including US variables in the specification. The following econometric exercise augments the benchmark econometric specification in Eq. (4) with a set of US control variables. This alternative econometric specification becomes

$$y_{i,t+h} = \beta_h^{MP} i_t^{\mathrm{MP}} + \beta_h^{RN} i_t^{\mathrm{RN}} + \\ + \sum_{j=1}^{J_y} \delta_i^j y_{i,t-j} + \sum_{j=1}^{J_x} \alpha_i^j x_{i,t-j} + \sum_{j=1}^{J_i} \left(\phi_i^j i_{t-j}^{\mathrm{MP}} + \varphi_i^j i_{t-j}^{\mathrm{RN}} \right) + \sum_{s=i}^{J_{US}} \alpha_i^{US} x_{US,t-j} + \epsilon_{i,t} \tag{7}$$

where $x_{US,t-j}$ represents a vector of US control variables.[35] Figures 79 and 80 in Appendix C show that the main results are robust to this specification, with slightly larger impacts and improved statistical significance.

4.2.2. Results across different sub-samples

Results across exchange rate regimes. To assess whether the results in Section 3 are driven by differences in exchange rate regimes, I classify countries into four groups based on Ilzetzki et al. (2017)'s coarse classification, ranging from 1 ("de facto peg" to the US dollar) to 6 ("dual market where parallel market data is missing"). Given that only 10% of observations fall under classifications 4 ("freely floating"), 5 ("freely falling"), and 6 ("dual market"), these are combined into a single group.

I classify countries according to their median classification over their sample. Figures 81 and 84 in Appendix C present the results. Across all groups, the MP component consistently leads to exchange rate depreciation, higher long-term rates, and a decline in industrial production and equity indices. In contrast, the RN component leads to exchange rate appreciation, no initial impact on long-term rates, and expansions in industrial production and equity indices. Following the "Standard HFI" approach produces an expansion in industrial production and equity indices for all groups. The magnitude of nominal exchange rate responses is lower for countries classified under a "de facto peg". These findings support the robustness of the main results across exchange rate regimes.

Results across levels of openness of the current account. Countries differ significantly in their degree of openness to cross-border financial transactions (Chinn and Ito, 2008), which may affect the transmission of US interest rate shocks.[36] To test for heterogeneity in spillovers, I classify countries into quartiles based on their mean current account openness index (Chinn and Ito, 2008). I then estimate Eq. (4) for each group.

Figures 85 to 88 in Appendix C present the impulse response functions for these groups. Across all quartiles, the MP component generates opposing spillovers compared to the RN component, consistent with the results in Section 3. Thus, current account openness does not drive the main results.

Results across levels of GDP per capita. I extend the AE and EME analysis (Fig. 4) by grouping countries into GDP per capita quartiles based on 2004 World Bank data, the midpoint of the sample period. Average GDP per capita across quartiles is $7,848, $15,907, $27,697, and $40,106 (constant 2015 USD). This analysis accounts for heterogeneity in income levels, which could influence international spillovers.[37]

Figures 89 to 92 in Appendix C confirm that results are consistent across all groups. The MP component leads to depreciation, higher long-term rates, and declines in industrial production and equity indices, while the RN component produces opposing effects.

Results across trade openness. Given the importance of trade in the transmission of US monetary policy (Camara et al., 2024), I test whether the results are influenced by countries' degrees of trade openness. Countries are classified into quartiles based on their average trade-to-GDP ratios, which range from 39.6% to 140.3%.[38] Figures 93 to 96 in Appendix C present the results. Across all quartiles, the MP and RN components yield opposite spillovers, consistent with Section 3.

Results across continents. I classify countries into Africa, Asia, Europe, Latin America, North America, and Oceania and estimate impulse response functions for each continent. Figures 97 through 102 in Appendix C show that the MP component leads to

[32] Previous versions imposed an ordering assumption in country-specific SVARs, which assumed uncorrelated structural shocks across countries.

[33] Figures 70 to 72 in Appendix C show that results are robust to estimating these impulse response functions for the different time-samples when replacing the nominal exchange rate with the trade-weighted real exchange rate.

[34] Figures 76 to 78 in Appendix C show results are robust when replacing the nominal exchange rate with the trade-weighted real exchange rate.

[35] In particular, I include the "Market Yield on U.S. Treasury Securities at 2-Year Constant Maturity, Quoted on an Investment Basis", the real dollar index constructed by the Federal Reserve, the PCE price index, the industrial production index constructed by the Federal Reserve, the US' total export and imports in current US dollars and the excess bond premium (see Gilchrist and Zakrajšek (2012)). As in Section 3, I include 2 lags of these US variables in the estimation.

[36] For instance, see Aizenman (1999).

[37] Data is sourced from the World Bank's dataset.

[38] Furthermore, there is evidence that countries that are relatively more open to trade experience less frequent and less severe Sudden Stop crises, see Cavallo and Frankel (2008).

depreciation and declines in industrial production and equity indices, while the RN component generates opposite effects. Following the "Standard HFI" approach yields muted exchange rate responses and expansions in industrial production. These results confirm that the findings are consistent across continents, with only minor quantitative differences.

Results using (Ilzetzki and Jin, 2021) dataset. To ensure the findings are not specific to my benchmark sample, I replicate the analysis using the dataset from Ilzetzki and Jin (2021).[39] I estimate the impulse response functions by estimating Eqs. (4)–(6).[40]

Figures 103 to 105 show that MP component shocks lead to exchange rate depreciation and industrial production declines, while RN component shocks lead to exchange rate appreciation and industrial production expansions. These patterns align with conventional results once the "Fed Response to News" channel is accounted for. Results remain robust across specifications and time samples, as shown in Figures 106 to 111.

Summary. These sub-sample robustness checks confirm that the main results in Section 3 are robust across varying exchange rate regimes, degrees of current account and trade openness, GDP per capita levels, geographic regions, and datasets. The findings emphasize the need to account for the "Fed Response to News" to estimate the international spillovers of US interest rates accurately.

5. Conclusions

This paper shows that the international spillovers of US interest rate hikes hinge on the type of underlying shock. Using the identification strategy proposed by Bauer and Swanson (2023a), I decompose monetary policy surprises into two components: a pure monetary policy shock (MP) and a "Fed Response to News" shock (RN). Incorporating these components into a panel of 50 countries spanning Advanced Economies and Emerging Markets, I find that the two shocks generate qualitatively opposite spillovers.

On the one hand, an MP shock leads to a nominal exchange rate depreciation, tighter financial conditions, and a persistent contraction in industrial output. On the other hand, an RN shock causes a nominal exchange rate appreciation, looser financial conditions, and a sustained expansion in industrial output. These findings underscore the importance of differentiating between the effects of monetary policy tightening and the systematic responses to economic news embedded in FOMC announcements.

I also demonstrate that following the standard high-frequency identification approach to US monetary policy leads to atypical spillover dynamics. This approach fails to account for the "Fed Response to News" channel, resulting in impulse response functions that reflect an average of the effects from MP and RN shocks. Under this strategy, a US interest rate hike induces unconventional outcomes, such as a significant expansion in industrial production and equity indices. Furthermore, by ignoring the RN channel, the standard HFI approach underestimates the quantitative impact of US monetary policy shocks on variables that do not display these atypical dynamics.

Finally, I show that the main results are robust across various models and sample specifications. The findings hold when analyzing Advanced and Emerging Market economies separately, using alternative local projection methodologies and SVAR models, considering different variable selections, and across variations in GDP per capita, exchange rate regimes, and current account openness. The results are also robust across geographic regions and using datasets from other studies that identified recent atypical dynamics. Notably, I confirm the robustness of the findings across three alternative identification strategies, each controlling for the RN channel with different methodologies and interpretations.

These findings highlight the critical role of the "Fed Response to News" channel in accurately estimating the international spillovers of US interest rates. Ignoring this channel risks drawing inaccurate conclusions about the effects of US interest rate changes on the world economy.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.jinteco.2025.104059.

[39] The dataset can be downloaded from the authors' website.

[40] Note that this last specification is the empirical specification used in Ilzetzki and Jin (2021).

Data availability

Spillovers of US Interest Rates (Original data) (Mendeley Data).

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